

IOT 2025/2026 - Homework

Exercise 1

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Data

$R = 250 \text{ kb/s}$ (nominal bitrate)

$L = 128 \text{ B} = 1 \text{ kb}$ (packet size)

$\lambda = 0.4$ (expected value of peaks inside a window)

$BI = 10 \text{ s}$ (beacon interval of the system = window duration)

$M_0 = 512 \text{ B} = 4 \text{ kb}$ (message payload size in case of 0 peaks in a window)

$M_1 = 2 \text{ kb} = 16 \text{ kb}$ (message payload size in case of 1 peak in a window)

$M_2 = 4 \text{ kb} = 32 \text{ kb}$ (message payload size in case of 2 or more peaks in a window)

CONVENTION USED: for the size of the packets $1 \text{ kb} = 1024 \text{ b}$ while for transmission rates $1 \text{ kb/s} = 1000 \text{ b/s}$.

Solution

First of all, I compute all the possible output rates for a single node:

- $r_0 = \frac{M_0}{BI} = 409.6 \text{ b/s}$
- $r_1 = \frac{M_1}{BI} = 1638.4 \text{ b/s}$
- $r_2 = \frac{M_2}{BI} = 3276.8 \text{ b/s}$

The Poisson distribution has the following probability mass function: $P(x=n) = \frac{\lambda^n}{n!} \cdot e^{-\lambda}$, where in our case x is the number of peaks detected in a window.

So the probabilities associated to the output rate r are computed as following:

- $P(r=r_0) = P(x=0) = \frac{\lambda^0}{0!} \cdot e^{-0.4} = \mathbf{0.670}$
- $P(r=r_1) = P(x=1) = \frac{\lambda^1}{1!} \cdot e^{-0.4} = \mathbf{0.268}$
- $P(r=r_2) = 1 - P(r=r_0) - P(r=r_1) = \mathbf{0.062}$

Then, I compute how many packets are needed in the three cases:

- 0 peaks detected: $\frac{M_0}{L} = 4$ packets
- 1 peak detected: $\frac{M_1}{L} = 16$ packets
- 2 or more peaks detected: $\frac{M_2}{L} = 32$ packets

In order to be able to determine a consistent CFP slots assignment, I consider the worst case scenario: 32 slots must be assigned to each device. Given that we have 3 nodes linked to the PAN Coordinator, **the total number of slots in the CFP is $N = 32 \cdot 3 = 96$.**

The time taken by a single slot is $T_s = \frac{L}{R} = 4.096 \text{ ms}$.

The total active time is $T_{active} = (N+1) \cdot T_s = 397.3 \text{ ms}$.

The inactive time is $T_{inactive} = BI - T_{active} = 9.603 \text{ s}$.

The duty cycle is $DC = \frac{T_{active}}{BI} = 3.973 \%$.

If other nodes are added and I want to keep $DC < 10 \%$, I get: the following condition about the maximum number of slots (N_{new}): $T_{active} < 0.1 \cdot BI \Rightarrow (N_{new} + 1) \cdot T_s < 0.1 \cdot BI \Rightarrow N_{new} < \frac{0.1 \cdot BI}{T_s} - 1 = 243.1$.

So the maximum number of devices that can be linked to the PAN coordinator is $\left\lfloor \frac{243.1}{32} \right\rfloor = 7$ which leads me to the conclusion that **only 4 other nodes can be added**.